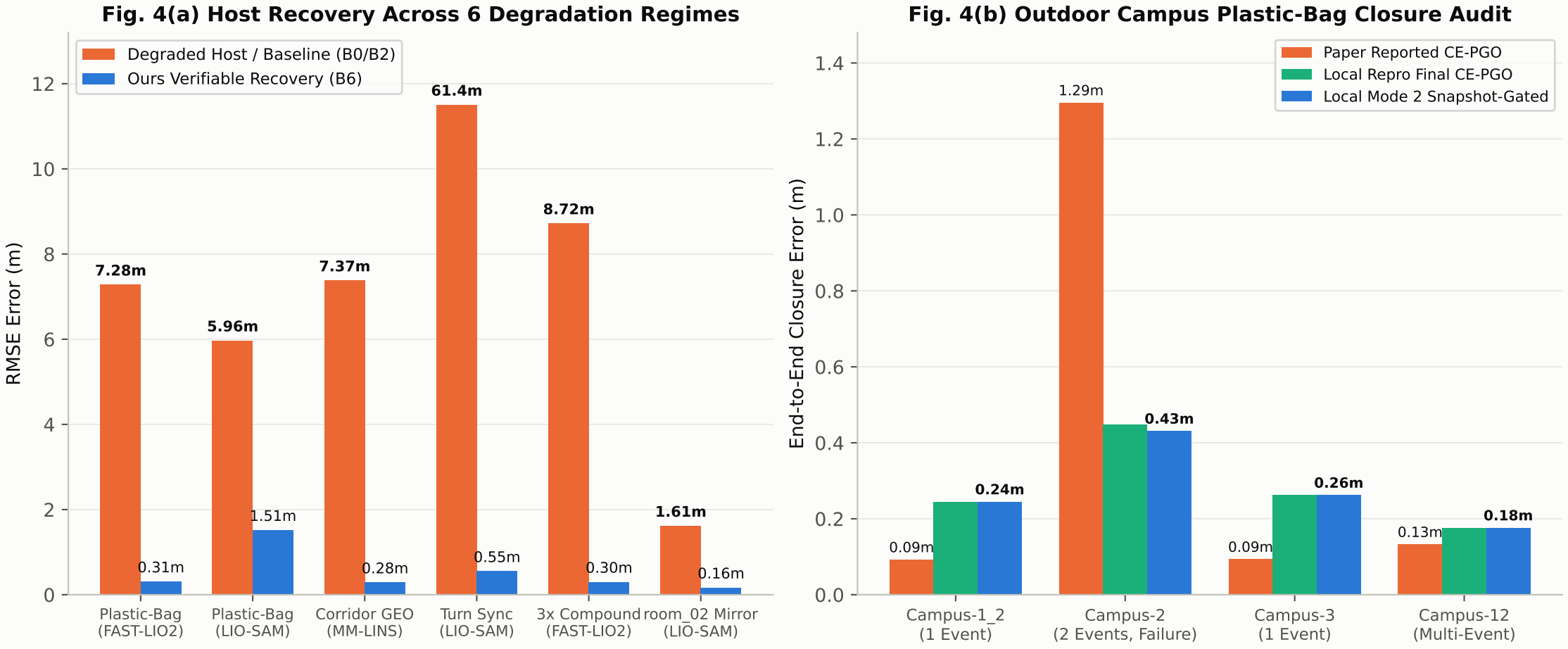
Fig. 4(a) Host Recovery Across 6 Degradation Regimes
Degraded Host / Baseline (B0/B2)
Ours Verifiable Recovery (B6)
RMSE Error (m)
0
2
4
6
8
10
12
7.28m
0.31m
5.96m
1.51m
7.37m
0.28m
61.4m
0.55m
8.72m
0.30m
1.61m
0.16m
Plastic-Bag (FAST-LIO2)
Plastic-Bag (LIO-SAM)
Corridor GEO (MM-LINS)
Turn Sync (LIO-SAM)
3x Compound (FAST-LIO2)
room_02 Mirror (LIO-SAM)
Fig. 4(b) Outdoor Campus Plastic-Bag Closure Audit
Paper Reported CE-PGO
Local Repro Final CE-PGO
Local Mode 2 Snapshot-Gated
End-to-End Closure Error (m)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0.09m
0.24m
1.29m
0.43m
0.09m
0.26m
0.13m
0.18m
Campus-1_2 (1 Event)
Campus-2 (2 Events, Failure)
Campus-3 (1 Event)
Campus-12 (Multi-Event)